

Too Much Information? Investigating Information Disclosure in Auction Systems with LLM Simulations

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Abstract

Online advertising relies on auction systems to allocate advertisement impressions efficiently and maximize revenue. A defining characteristic of these systems is information asymmetry: publishers (auctioneers) have access to detailed data, while bidders rely on limited signals. Understanding how information disclosure strategies shape bidder behavior and auction outcomes is essential for designing adaptive, user-centered advertising systems.

This paper introduces **InfoBid**, a simulation framework powered by large language models (LLMs) to study the impact of information signaling in auction environments. This simulation framework enables researchers to analyze how information disclosure affects decision-making and system performance. By offering a flexible and extensible testbed, InfoBid advances Human-Computer Interaction (HCI) methodologies for studying multi-agent interactions, adaptive decision-making, and information sharing in complex digital marketplaces.

CCS Concepts

• **Computing methodologies** → **Simulation environments**;
• **Multi-agent systems**; • **Applied computing** → **Economics**; •
• **Human-centered computing** → **Interactive systems and tools**.

Keywords

Multi-Agent Simulation, Information Asymmetry, Strategic Information Disclosure, Large Language Models (LLMs), Mechanism Design, Digital Advertising System

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1 Introduction

Display advertising plays a crucial role in the digital economy, generating billions of dollars each year. Platforms like Google and Meta (publishers) sell user impressions to advertisers such as Coca-Cola, Amazon, and Nike through real-time auctions. These advertising

systems involve complex, multi-agent interactions: advertisers (bidders) plan and submit bids for impressions, publishers (auctioneers) allocate impressions, collect payments, and refine the auction mechanism based on auction outcomes, while users engage with the impressions delivered through these auctions. Research in HCI on advertising systems has highlighted the importance of transparency, usability, and fairness, providing valuable insights into how system design can improve user engagement and support decision-making in these complex, AI-driven environments [3, 5, 8, 9].

Auction systems are central to advertising systems, efficiently allocating resources in real time and forming the foundation of many digital marketplaces. Despite their key role in shaping user interactions, economic outcomes, and platform design, auction systems remain less explored in HCI compared to AI-driven systems such as recommendation algorithms or social media platforms. One of the main challenges in designing auction systems is the information asymmetry between auctioneers and bidders. Platforms like Google and Meta use extensive datasets on user behavior—such as browsing history, click patterns, and purchase intent—combined with detailed advertiser data, including bidding histories and campaign performance. In contrast, bidders often have access to limited, aggregated, or anonymized information. For example, Nike might only know that their target audience consists of fitness enthusiasts aged 18–35, without detailed user-level data or real-time updates. This imbalance forces bidders to make decisions with incomplete information, leading to inefficiencies and missed opportunities.

This asymmetry complicates auction system design, as it creates challenges in ensuring efficient resource allocation while balancing the different objectives of bidders and auctioneers. Information disclosure has emerged as an important strategy to address this issue. For instance, platforms may share targeting keywords such as “luxury vehicles” or “organic food” to give bidders hints about user information or offer auto-bidding tools that adjust bids dynamically based on user data. While such disclosure strategies can improve decision-making, they also come with trade-offs: too much detailed information may reduce competition, while insufficient disclosure can lead to inefficiencies and dissatisfaction among participants.

The challenges of appropriate information disclosure extend beyond advertising systems to broader areas in HCI, such as social media, privacy management, and recommender systems [12, 16, 20, 21, 27, 28]. Decisions about what information to disclose can significantly influence user behavior, engagement, and decision-making across various contexts. Building on these themes, this research explores a relatively under-explored area: *how information disclosure strategies affect decision-making and outcomes in multi-agent auction systems*. By focusing on auction systems,

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this study addresses a critical gap in HCI research, expanding the understanding of information design in AI-mediated interactions.

Simulations are a powerful method for studying these dynamics in a controlled, risk-free environment. They allow researchers to test hypotheses and evaluate system designs without operational or ethical risks, bridging the gap between theoretical design and practical application. Their effectiveness has been demonstrated in various industry applications [10, 11, 13, 29]. This paper introduces **InfoBid**, a simulation framework powered by Large Language Model (LLM) agents. InfoBid is designed to address the limitations of existing simulation frameworks, which often rely on rigid, rule-based models that struggle to capture human decision-making adaptability and lack interpretability. LLM agents, such as GPT-4, simulate human-like reasoning, adapt to complex scenarios, and provide interpretable explanations for their actions [6, 14, 23]. These capabilities have been validated in HCI research [1, 15, 17–19] but remain largely unexplored in the context of auction systems, especially for studying information disclosure. By integrating insights from HCI and auction theory, InfoBid serves as a novel platform for understanding how information design impacts multi-agent decision-making. It offers valuable insights for designing efficient, adaptable, and scalable systems in digital marketplaces.

2 Methodology

The InfoBid framework is a simulation framework designed to explore how information disclosure strategies influence agent behavior and outcome in auction environment. It supports configurable auction mechanism, such as first-price and second-price auctions with or without reserve prices, as well as diverse information disclosure strategies, allowing for flexible and extensible evaluations. InfoBid comprises three key components: (1) **Agents**, including the auctioneer (who manages the auction) and multiple bidders (who compete to win); (2) **Information Structure**, which defines what information is shared, hidden, or pooled to shape participant decision-making; (3) **Auction and Signaling Process**, which simulates the flow of information, bid submissions, and the determination of winners.

2.1 Agent Design

2.1.1 Auctioneer. The auctioneer plays a central role in the framework, managing the auction and typically possessing more information than the bidders. Two key settings are widely studied in the literature [4]: *Known-Valuations*, where the auctioneer knows each bidder's exact valuation for the auctioned item, and *Bayesian Setting*, where the auctioneer operates with a "prior belief" based on probabilistic estimates rather than precise valuations. While our framework can support both settings, this study focuses on the Known-Valuation setting, where the auctioneer has complete knowledge of each bidder's true valuation. This reflects real-world scenarios in digital advertising, where online platforms leverage extensive data to accurately estimate user preference and how much each bidder values a user impression.

2.1.2 Bidder. Bidder agents represent the participants in the auction and are tasked with formulating strategic bids based on the signals received from the auctioneer and their own knowledge. Unlike the auctioneer, bidders lack information about the auctioned

item and are unaware of their true valuations. In this study, we use GPT-4o as the reasoning engine for bidders, with the temperature set to 0 to ensure reproducibility. Bidders follow a two-step process: (1) estimate their valuations using the provided signals and their knowledge—for instance, a signal like "*You are among the high-value bidders*" might lead a bidder to infer a relatively high valuation while remaining uncertain about its exact value; and (2) formulate bids based on these estimates. For example, a bidder estimating their valuation at 0.75 might bid slightly lower to account for uncertainty while remaining competitive. Additionally, bidders are required to explain their bidding decisions, providing insights into their reasoning process.

2.2 Information Structure

The InfoBid framework distinguishes between *Public Information* and *Private Information* to model information disclosure strategies:

- **Public Information:** Shared among all agents, this serves as a common reference for decision-making and may include:
 - Auction Rules: Details of the auction mechanism (e.g., first-price or second-price), number of bidders, and tie-breaking rules.
 - Common Prior on Bidders' Distribution: For example, bidders and the auctioneer may agree that valuations follow a uniform distribution or depend on bidders' type;
 - Signaling Map: Describes how each bidder's valuation is transformed into specific signals. For example, the signaling map might suggest that high-value bidders (e.g., those above the 80th percentile) will have their information pooled, while low-value bidders will have their valuations fully disclosed.
 - Auction outcome: After each round, the auctioneer reveals results, including the winning bidder and the price paid.
- **Private Information:** Exclusive to each bidder, this primarily includes their valuation of the auctioned item in our framework. The auctioneer may disclose private information using:
 - Full Disclosure: Revealing a bidder's exact valuation (e.g., "Your valuation is 0.7"), removing uncertainty and encouraging truthful bidding.
 - Selective pooling: Aggregating and sharing approximate signals with specific bidder groups.

2.3 Auction and Signaling Process

InfoBid employs a structured six-phase pipeline, from initializing the auction context to announce the auction outcome, to simulate the auction and signaling process, providing a comprehensive approach to analyzing the effects of information disclosure strategies on auction outcomes and bidder behavior. Figure 1 illustrates the information flow between the auctioneer and bidders in an example auction. The auctioneer agent module has full visibility into the auction context, including item characteristics, which determine the exact valuation for each bidder. Based on this information, the auctioneer applies a predefined information disclosure strategy, deciding what details to reveal or withhold from bidders. The bidding agents module presents three types of information from left to right. On the left, private messages contain information provided by the

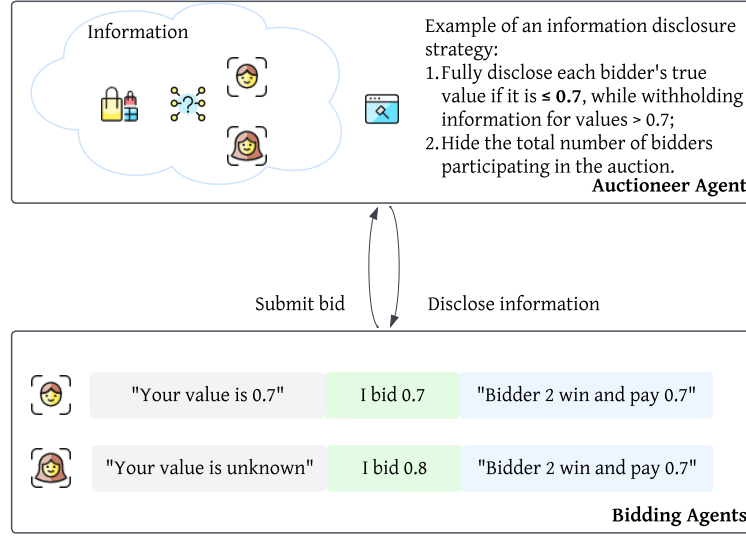


Figure 1: Information flow between auctioneer and bidders in an example auction.

auctioneer to individual bidders. In the middle, sealed bids are submitted privately by bidders to the auctioneer. On the right, public messages display information accessible to all participants. This setup illustrates a specific disclosure strategy, defining conditions for revealing true values while withholding the total bidder count. Detailed introduction of the phases is in Appendix.

3 Experiments

3.1 Simulation Design

We conducted simulations using a sealed-bid second-price auction, a widely studied mechanism in both theory and practice [25]. In this mechanism, bidders submit private bids without knowing others' bids. The highest bidder wins the item but pays the second-highest bid. This format incentivizes bidders to bid truthfully according to their perceived valuations, providing a robust framework to analyze how various information disclosure strategies influence strategic decision-making. To simplify the setup, we include 10 bidders in the simulation, with each bidder's valuation independently drawn from a uniform distribution over $[0, 1]$.

The default public information structure includes: (1) The auction type (sealed-bid second-price) and the number of bidders; (2) The common prior that all bidders' valuations for each item are independently drawn from a uniform distribution over $[0, 1]$. Notably, the signaling map used to disclose private information is not shared publicly, meaning bidders are unaware of the information their competitors receive.

The simulation systematically varied private signaling strategies to evaluate their impact on bidder behavior and auction outcomes. The tested strategies include (For further details on the prompts used in each scenario, refer to the Appendix):

- Full disclosure: Bidders are informed of their exact valuation.

- Tiered pooling ("Pool-High" and "Pool-Low"): Bidders are grouped into quantile-based tiers by auctioneer based on their true-values. For instance, the "Pool-Hig" strategy provides pooled information to bidders whose true valuations fall within the higher percentiles while fully disclosing valuations to the remaining bidders (See appendix for the formula). Similarly, the Pool-Low strategy applies pooling to lower-valued bidders, selectively revealing their positional context. Tiered pooling sub-cludes two variants of pooled information: 1) Tier-only information, where bidders are informed solely of their relative position within a tier, e.g., "You are among the high tier of bidders."; 2) Tier and average value information, where bidders receive their relative position along with the average true valuation within their tier, e.g., "You are among the high tier of bidders, and the average valuation in your tier is 0.8."
- Randomized pooling: Bidders are randomly selected to receive full disclosure, while others are provided pooled information based on predefined probabilities. Pooled bidders receive no average-value information.

Both tiered pooling and randomized pooling are parameterized using probabilistic thresholds to control the proportion of bidders receiving pooled versus fully disclosed information, enabling direct comparisons across strategies.

Each configuration was executed for 100 independent rounds to ensure robust findings. In every round, a new item was introduced, and bidders were assigned fresh valuations drawn from the uniform distribution, ensuring independent and identically distributed (i.i.d.) conditions. For simplicity, the framework treated each round as an isolated decision-making snapshot, without incorporating iterative learning across rounds. See Appendix for the full list of simulation configurations and prompts used in each scenario.

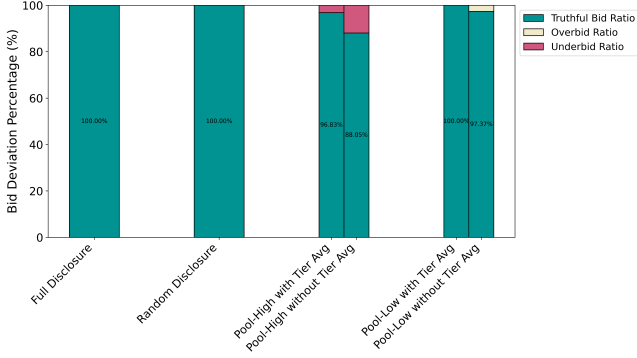


Figure 2: Impact of Signaling Strategies on Bid Deviation Behavior

3.2 Evaluation Metrics

We used the following metrics to evaluate the effectiveness of information disclosure strategies:

- **Bid Deviation from Rational Behavior:** In second-price auctions, rational bidders are expected to bid truthfully, submitting bids equal to their estimated true valuation. This metric measures how closely submitted bids align with the bidders' perceived valuations. Deviation is calculated as the percentage of bids that are greater than, equal to, or less than their estimated values, offering insights into deviations from theoretically optimal behavior.
- **Revenue:** Revenue represents the total amount paid by winning bidders across all rounds. This metric evaluates the auctioneer's performance under different disclosure strategies and highlights how information sharing influences competitive bidding.
- **Social Welfare:** Social welfare measures the total valuation of winning bidders over all rounds, assessing how efficiently the auction allocates items to those who value them the most. It is defined as:

$$SocialWelfare = \sum_{r=1}^R v_r$$

where R is the total number of auction rounds, and v_r represents the true valuation of the winning bidder in round r . This metric evaluates the overall efficiency of the auction in allocating items to bidders with the highest valuations.

3.3 Results

3.3.1 Bidder's Rationality Analysis. We analyze bidder behavior under different information disclosure strategies, focusing on how bidders infer their valuations and make decisions in response to provided signals. Rational bidding behavior is defined as submitting bids that align with the bidder's inferred valuation, with theoretical expectations suggesting truthful bidding in second-price auctions. Figure 2 illustrates deviations from this behavior across disclosure strategies. We observe consistent rational bidding behavior under full disclosure and randomized pooling, where LLM-based agents submitted truthful bids, aligning with theoretical expectations in

second-price auctions. This outcome demonstrates the logical reasoning capabilities of LLM-based agents when provided with clear and unambiguous signals.

Tiered pooling strategies introduced notable deviations from truthful bidding. Under the Pool-High strategy, high-value bidders tended to underbid, likely due to heightened competition awareness within their tier. Conversely, the Pool-Low strategy prompted overbidding among low-value bidders, who perceived greater opportunities to compete aggressively. These behaviors align with social comparison theory, which suggests that individuals adjust their decisions based on their perceived position relative to others [7, 22, 26]. Providing additional information, such as the average valuation within a tier, mitigated these deviations by instilling greater confidence in estimated values. This behavior reflects cognitive biases like the anchoring effect [24], where agents adjust bids based on reference points such as disclosed average valuations. For example, bidders informed that their tier's average valuation is 0.66 often bid near this value (see Appendix for bidder explanations).

Interestingly, while LLM-based agents provided detailed explanations for their decisions, these explanations lacked explicit reasoning about competitors' potential actions or values, suggesting limited strategic reasoning about competition. This aligns with theoretical expectations for second-price auctions, where truthful bidding is a dominant strategy independent of competitor behavior. However, it remains unclear whether this behavior is a result of LLM rationality or limitations in their reasoning capabilities. Exploring first-price auctions, where competitive reasoning is crucial, could provide deeper insights into LLM agents' decision-making and adaptability under different auction formats, offering an avenue for future research.

3.3.2 Revenue and Social Welfare Analysis. The results highlight how different information disclosure strategies affect both auction revenue and allocation efficiency. Figure 3 illustrates revenue outcomes, while Figure 4 shows social welfare, defined as the total valuation of items won by bidders, reflecting allocation efficiency. The charts illustrate the relationship between the probability of full disclosure and average performance per round. The x-axis represents the average probability of full disclosure across different scenarios—for example, pooling bidders with values above the 20th percentile corresponds to a full disclosure probability of 0.2, while a random disclosure rate of 50% corresponds to a full disclosure probability of 0.5. The y-axis denotes the average performance (revenue or social welfare) per round generated under each configuration.

Revenue Insights. Both the Pool-Low strategy and Full Disclosure consistently generate similar revenue outcomes, as they ensure that high-value bidders receive precise and actionable information. This enables confident and truthful bidding, leading to predictable and efficient revenue generation for the auctioneer. Conversely, the Pool-High strategy demonstrates a clear trend of increasing revenue as the proportion of disclosed information rises. High-value bidders, provided with clearer signals such as tier averages, adjust their bids to more accurately reflect their valuations. This behavior, observed in the rationality analysis, underscores how enhanced bidder confidence through targeted information disclosure can incentivize more aggressive bidding, ultimately boosting revenue.

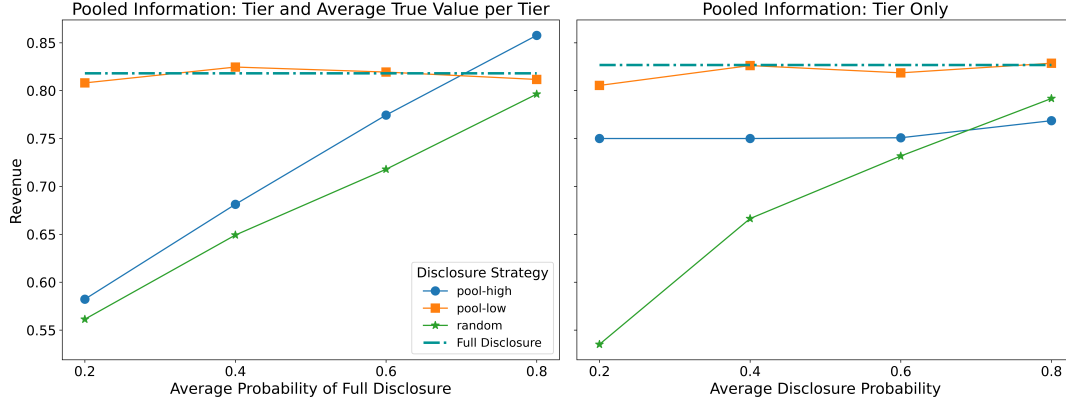


Figure 3: Revenue Comparison Across Disclosure Strategies and Scenarios.

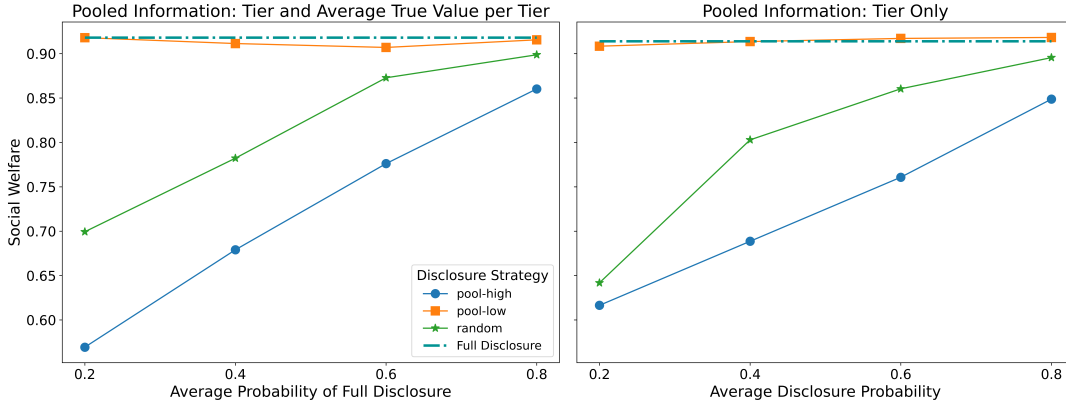


Figure 4: Social welfare Comparison Across Disclosure Strategies and Scenarios.

Interestingly, when pooled information is limited to tier-only details, high-value bidders often rely on heuristics, bidding around a stable constant value (the bidding records in our study suggest 0.75, which happens to be the average value from a uniform distribution drawing $[0.5, 1]$), resulting in stable revenue across quantiles. In contrast, adding tier average values significantly increases revenue, as the additional detail creates significant information rents for top-tier bidders. For instance, pooling the top two bidders yields a higher average true value than pooling the top five bidders because the smaller subset is concentrated at the upper end of the value distribution. As we see from the rationality analysis, the higher average shapes bidder’s belief and incentivizes aggressive bidding among top-tier bidders, leading to revenue gains. These findings align with theoretical results from Bergemann et al., demonstrating the importance of balancing detailed feedback with competitive tension to optimize outcomes in system design.

Social Welfare Insights. Social welfare results reveal how disclosure strategies impact the auction system’s allocation efficiency. The Pool-High strategy initially underperforms compared to Full Disclosure and Pool-Low, as withholding critical information from high-value bidders reduces their ability to bid effectively. However, as more detailed signals are disclosed, the Pool-High strategy

improves, approaching Full Disclosure levels of efficiency. This demonstrates the value of increasing transparency to enable more confident decision-making among influential users.

Full Disclosure and Pool-Low consistently achieve higher social welfare by ensuring that high-value bidders—who contribute most to efficient allocation—receive sufficient information to make optimal bids. Notably, Pool-Low achieves stable social welfare across varying thresholds, indicating that limiting information for lower-value bidders does not compromise efficiency, provided that high-value bidders are well-informed. This highlights the importance of tailoring information strategies to prioritize users who have the greatest impact on outcomes.

4 Conclusion

This paper introduces InfoBid, a simulation framework powered by LLM agents, to investigate how information disclosure strategies influence bidder behavior and system outcomes in auction environments. By simulating decisions under varying information conditions, InfoBid illustrates how transparency and information design impact user decision-making, system efficiency, and economic outcomes. The results demonstrate that strategically crafted

disclosure strategies can balance revenue optimization with allocation efficiency, offering actionable insights for designing adaptive, user-centered auction systems.

As auction systems play an increasingly central role in digital platforms like online advertising and e-commerce, this research addresses critical themes in HCI, such as transparency, usability, and the design of AI-mediated interactions. By leveraging LLM-based agents, InfoBid provides a flexible and extensible testbed for studying multi-agent decision-making in complex systems. This approach aligns with HCI's mission to enhance user experiences in data-driven environments, showing how tailored information disclosure strategies can improve system performance and foster equitable outcomes.

The findings from InfoBid highlight essential considerations for designing auction systems: how to craft disclosure strategies that enhance user understanding while meeting platform objectives. These insights can guide platforms in building adaptive, data-informed systems that balance user engagement with economic goals. Future work could extend InfoBid to explore alternative auction formats, iterative interactions among agents, or real-world applications involving user-facing interfaces. By bridging AI-driven simulation with practical system design, this research contributes to advancing innovation in user-centered digital ecosystems.

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A Prompt Inputs to Bidder Agents

Information Structure

- (1) **Default public information:** You are a rational bidder in a second-price auction. The auction is one-time: you submit the bid and will receive the results. As a prior, you don't know your true value towards the current item. You only know your true-value falls between $[0, 1]$. Auctioneer may give you more information. The value of the item for each bidder is independently drawn from a uniform distribution over $[0, 1]$ and the auction will consist of 10 bidders, including you. This is common knowledge to all bidders. However, your exact value is private and known only to you, based on signals or disclosed information provided by the auctioneer.
- (2) **Private signaling:**
 - **Full Disclosure:** Your true value towards this current auctioned item is {current average value}
 - **Tiered Disclosure without Average Tier Value:** Your true value towards this current auctioned item is not disclosed due to your value is being in the {POOL LEVEL} value tier. This indicates your value towards this item is {POOL LEVEL}er than some of other bidders, but the exact value will remain unknown.
 - **Tiered Disclosure with Average Tier Value:** Your true value towards this current auctioned item is not disclosed due to your value is being in the {POOL LEVEL} value tier. This indicates your value towards this item is {POOL LEVEL}er than some of other bidders, but the exact value will remain unknown. The average value of all bidders in the same tier with you is {TIER AVG VALUE}.
 - **Randomized Pool:** You have no information about your true value.

Bidding Instruction

Based on this information, submit a bid between 0 and 1, and provide the following details in your response: name: your name as a bidder. bid: Your bid for this round (a value between 0 and 1). estimated value: Your estimate of your true value based on the information received. explanation: A brief explanation of how you decided at your bid. Please return your response in the provided format

B Bidder's Reasoning

Examples of Bidder's Explanation on Bidding Decision

- (1) **Explanation under Full Disclosure:** In a second-price auction, the optimal strategy is to bid your true value. This is because if you win, you will pay the second-highest bid, not your own. Therefore, bidding your true value maximizes your utility, ensuring you win the item if the second-highest bid is less than or equal to your true value. Given my true value is 0.3568638462861372, I will bid exactly this amount.
- (2) **Explanation under Tiered Pooling:**
 - **Bid truthfully on the disclosed average value:** Given that my true value is in the high-value tier and the average value of bidders in this tier is 0.6657625810811798, I estimate my true value to be around this average. In a second-price auction, the optimal strategy is to bid your true value. Therefore, I bid 0.6657625810811798, aligning with my estimated true value.
 - **Bid apart from the estimated value :** Given that my value is in the high value tier and the average value of this tier is 0.6610732108060862, I estimate my true value to be slightly above this average, at around 0.7. In a second-price auction, the optimal strategy is to bid your true value. However, since I don't know my exact value, I will bid the average of my tier, 0.6610732108060862, as it is a reasonable estimate of my true value.
- (3) **Explanation under Randomized Pooling:**
 - **Bid without any information disclosed:** Without specific information about my true value, I assume the expected value of my true value is the midpoint of the distribution, which is 0.5. In a second-price auction, the optimal strategy is to bid your true value. Therefore, I bid 0.5, which is my estimated value based on the uniform distribution assumption.

C Simulation Variants

Table 1 in illustrates the specific configurations in our simulation design.

D Auction and Signaling Process

In this section, we introduce the detailed auction and signaling process used in our framework:

- **Auction Context Initialization:** In this initial phase, the auctioneer gathers all relevant information for the current auction round.

Factor	Variations
Number of Bidders	10
Value Disclosure Strategy	Full Disclosure, Pool-High, Pool-Low, Randomized Pooling
Threshold Quantiles	0.2, 0.4, 0.6, 0.8
Pooled Information for Tiered Pooling	Tier Only, Tier and Average Value Information
Repetitions per Configuration	100

Table 1: Simulation Variations

- **Information Disclosure:** The auctioneer discloses information to bidders according to the pre-defined strategy. This process includes public information, such as the number of participants or common priors, and private signals tailored to each bidder.
- **Signal Reception and Belief Update:** Upon receiving the disclosed information, bidders analyze received information to update their beliefs about their true valuations and adapt bidding decision strategically. This process is operationalized through LLMs.
- **Bid Generation and Submission:** Guided by their updated beliefs, bidders generate their bids for the current auction round. This process is operationalized also through LLMs.
- **Bid Collection and Auction Execution:** The auctioneer collects all submitted bids and runs the auction based on the configured mechanism.

- **Outcome Announcement** After executing the auction, the auctioneer publicly announces the results, potentially including the winner's identity and the price paid.

E Mathematical Formula Describing the Pooling Strategies

The mathematical formulation of "Pool-High" is shown in Eq. 1, where $F(v_i)$ is the cumulative distribution function (CDF) of v_i , q is the quantile threshold, and $p(v_i)$ represents the pooled information.

$$s(i) = \begin{cases} v_i & F(v_i) < q, \\ p(v_i) & F(v_i) \geq q \end{cases} \quad (1)$$

The mathematical formulation of "Randomized Pooling" is shown in Eq. 2:

$$s(i) = \begin{cases} v_i & \text{with probability } q, \\ p(v_i) & \text{with probability } 1 - q, \end{cases} \quad (2)$$